

CLARITY: Predicting Gen-AI Impacts and Optimizing Institution Programs

Summary

Against the backdrop of generative AI deeply embedding itself into learning and workflows, different occupations exhibit significant heterogeneity in task substitutability, skill restructuring, and benefit distribution. The education system urgently needs to identify key occupations, predict their development trajectories, and translate these findings into actionable recommendations for optimizing training programs and governance all through a data-driven approach. To achieve these objectives, we propose the CLARITY (CLustering Assessment for foRecast optImizaTion sYstem) model. During data preprocessing, the model constructs an occupation-year panel dataset from four authoritative public databases (BLS, NCES, College Scorecard, O*NET), obtaining comparable samples by handling missing values and standardizing variables. In CLARITY's occupation identification phase, Submodel I employs Entropy-Weighted Clustering (EWC): It constructs a four-dimensional indicator system covering labor demand, economic returns, skill structure, and technology dependency. Using entropy-weighted clustering, it generates annual composite score trajectories and identifies occupational types via K-means clustering. Ultimately, it selects railway maintenance workers, management analysts, and graphic designers as representative anchor occupations for Trade, STEM, and Arts, respectively, for subsequent impact assessment and scenario forecasting.

In CLARITY's driver analysis and forecasting stage, Submodel II employs WDT (Weighted Diffusion Timecasting): At the macro level, a Logistic diffusion impact function is introduced, with exponential smoothing estimating baseline trends in a non-AI scenario. This generates AI-adjusted future trajectory projections, enabling comparative trend analysis across the three representative occupations.

Finally, in CLARITY's institutional decision phase, Submodel III employs PMA (PSO Multi-objective Allocation): For three types of institutional projects, it constructs a six-objective planning model encompassing employability, scale matching, learning efficiency, knowledge accessibility and resource utilization, ethical and originality risks, and resource conservation. Under variable boundaries, proportional constraints, and budget constraints, it employs a particle swarm algorithm to determine optimal curriculum structures and governance investment allocations, thereby generating categorized, actionable policy-combination recommendations.

Sensitivity analysis demonstrates that the CLARITY model can provide tailored recommendations for institutions with different objective orientations, indicating its strong generalizability.

Keywords: Gen-AI; Entropy-Weighted Method; K-means; Time Series Analysis; Fuzzy Comprehensive Evaluation; Multi-Objective PSO.

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1 Introduction

1.1 Problem Background

In recent years, generative artificial intelligence has become deeply embedded in learning and work processes, providing support in writing, programming, design, information retrieval, and content generation, while reshaping workflows across various job roles. Employer surveys indicate that demand for key professional skills will change significantly in the coming years, making bridging the AI skills gap a critical challenge for businesses. The impact of AI development on the future of careers and the resulting transformation of talent development models have drawn widespread attention.

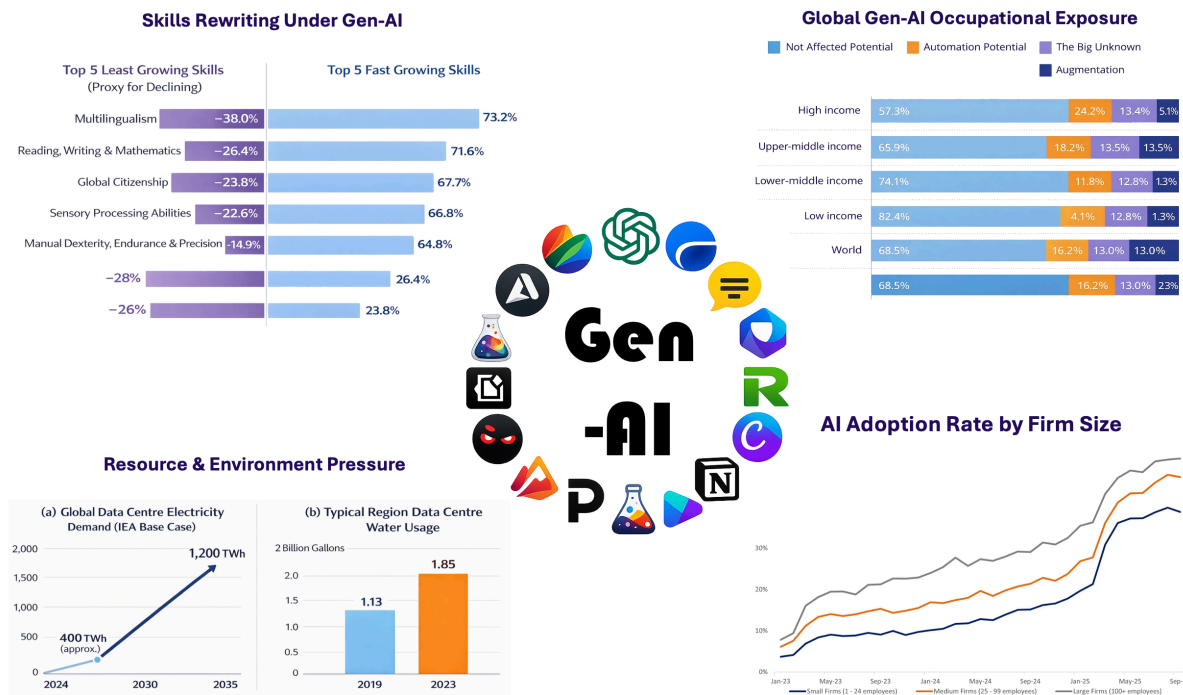


Figure 1: Gen-AI Impact Evidence. These charts synthesize multiple datasets[1, 2, 3, 4, 5] to illustrate the impact of Gen-AI on contemporary society and the environment.

As shown in figure 1, despite the widespread acceleration of AI diffusion across various fields, its impact on different occupations exhibits significant heterogeneity. While generative AI can support or partially complete numerous tasks in certain professions, others rely more heavily on on-site operations or interpersonal collaboration. Task-level estimates indicate that approximately one-quarter of the global workforce will experience varying degrees of impact. Common outcomes tend to be job restructuring and upskilling rather than direct job replacement. This projection creates an urgent imperative for education systems: curricula and assessment frameworks must be promptly adapted to equip students with the skills to leverage generative AI for capability enhancement while mitigating its negative consequences. Concurrently, the proliferation of generative AI imposes resource and environmental pressures. Model training and deployment consume substantial electricity, while hardware cooling increases water usage, impacting local water resources and

ecosystems. Therefore, there is an urgent need to employ data-driven methods to identify representative key occupations under AI penetration, systematically predict their development trajectories, and use these insights to inform educational model optimization and policy decision-making.

Against this backdrop, this study connects labor-market shifts to educational decision-making through data-driven analysis. The study first analyzes task evolution and skill demands across occupations, then translates these findings into actionable adjustments for schools and specific training programs. These adjustments encompass curriculum content, practical training, assessment design, faculty allocation, and risk management measures such as usage protocols, attribution, and acknowledgments. Through multi-indicator evaluation and uncertainty analysis, the research aims to provide more robust recommendations and construct a reusable decision-making framework applicable to diverse training programs.

1.2 Problem Restatement

Problem 1: Identify the most promising occupations for focused research across a wide range of professions, ensuring they represent key types of change and ultimately cover one occupation each from the three major categories: Trade, STEM, and Arts.

Problem 2: Develop a quantifiable model grounded in reliable data and prior research to predict both the direction and intensity of generative AIs impact on career development across the three categories, producing interpretable forecasts and trend assessments, while diagnosing the dominant driving factors that explain the predicted impacts.

Problem 3: Select one specific program each from trade schools, universities, and art institutions. Based on the preceding career trend analysis and Gen-AI impact assessment, formulate three sets of actionable recommendations for institutional leadership.

1.3 Our Work

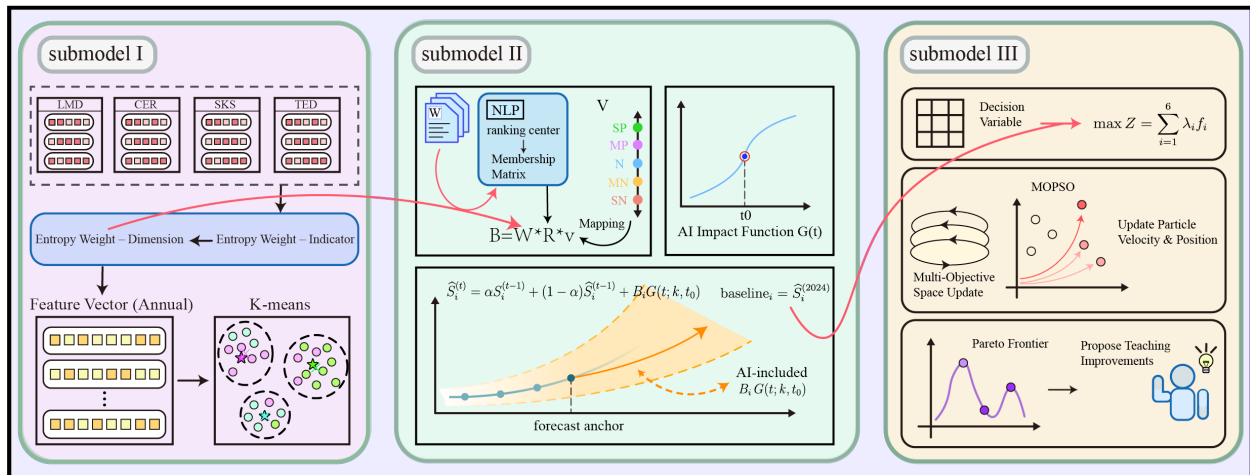


Figure 2: A two-dimensional t-SNE visualization of occupations colored by their occupational category labels (Trade, STEM, or Arts). The dashed circles in the figure represent clustering results.

We propose the CLARITY model. As shown in figure 2, first, it identifies career types and rep-

representative career anchor points through entropy-weighted composite scoring combined with K-means clustering. Then, it generates AI-adjusted career development trajectory predictions using NLP-assisted fuzzy comprehensive evaluation, logistic diffusion impact, and time series smoothing. Finally, a constrained multi-objective programming model is constructed. The Pareto frontier is solved using MOPSO, yielding actionable recommendations for course structures and governance investment policy combinations tailored to different types of institutions.

2 Assumptions and Justifications

Assumption 1: Data Accuracy

Justification: The model relies on authoritative public data sources, including BLS[6], NCES[7], CSD[8], and O*NET[9]. These data serve as foundational inputs for both occupational and educational dimensions. The model assumes consistent statistical methodologies and continuity in indicator definitions and collection methods across long-term database updates. It further assumes no future major institutional adjustments or statistical method changes that would cause systematic reconfigurations of these methodologies. Residual errors primarily manifest as random noise rather than systematic bias. This framework does not alter the conclusions drawn from relative comparisons across occupations and years.

Assumption 2: The indicator system is comprehensive and disregards the impact of indicators without metrics.

Justification: The model employs a multidimensional indicator system. The system covers core dimensions including labor demand, economic returns, skill structure, and technology dependency. The indicator system is used to characterize occupational differences and evolutionary patterns. This study assumes that excluded variables have a limited impact. These variables contribute marginally to the final comprehensive evaluation and optimization conclusions. They will not cause a systemic reversal in decision-making direction.

Assumption 3: The impact of generative AI will gradually intensify as it becomes more widespread, eventually stabilizing.

Justification: The development of new technologies typically unfolds in three distinct phases: adoption, diffusion, and maturity. This paper applies this framework to project the trajectory of generative AI's impact. Its influence on occupational tasks and skill requirements intensifies as adoption rates rise. Upon reaching maturity, the impact's intensity gradually stabilizes. This model enables future scenario analysis to capture temporal dynamics, avoiding the simplistic view of impact as a one-time, sudden transformation.

Assumption 4: Institutional reforms can be approximated by a small number of continuous proportional variables.

Justification: To ensure the model's operational feasibility, this paper describes training programs that use continuous variables, such as proportions of course modules and enrollment changes. It assumes that, within budgetary and structural constraints, the direction of these variables' effects on objectives, including employability, learning quality, inclusivity, resource utilization, and ethical compliance, remains stable. Simultaneously, the underlying mechanisms are interpretable. Conse-

quently, the multi-objective trade-off results can serve as a reference for policy and management-level resource allocation decisions.

3 Data Sources and Preprocessing

3.1 Data Sources

To ensure traceability and reproducibility of conclusions, as shown in Table 1, this paper uniformly uses four authoritative public databases as its foundational data sources. These databases comprehensively cover labor market outcomes, higher education supply and student outputs, and occupational-level task and skill profiles, thereby supporting subsequent clustering, comprehensive evaluation, and scenario analysis.

Labor Market and Compensation (BLS). The U.S. Bureau of Labor Statistics (BLS) provides occupational-level employment and wage statistics, which can be used to measure employment levels, compensation, and trends over time for occupations. It serves as the core source for the labor-demand and income-return indicators in this paper.

Institutional and Enrollment Statistics (NCES). The National Center for Education Statistics (NCES) provides institutional-level information and annual statistics for colleges, universities, and vocational institutions through systems such as IPEDS. This data characterizes institutional attributes and serves as a standardized key and background information source for linking education-end data.

Educational Outcomes and Returns (CSD). The U.S. Department of Education's College Scorecard database provides metrics at the institutional level and for select programs, including completion rates, debt and repayment, and post-graduation earnings. These metrics measure the economic returns and financial burdens of different educational pathways and support the assessment of education-to-career trajectories.

Occupational Tasks and Skills (O*NET). O*NET OnLine provides structured information at the occupational level, including task descriptions, skill and competency requirements, and technical skills. This paper utilizes this repository to construct occupational task and skill features, offering actionable proxy variables for characterizing occupations' potential exposure to generative artificial intelligence.

3.2 Preprocessing

To ensure comparability across different data sources and years, this study employs a two-step cleaning and missing value handling process for the original occupational data sample. First, occupations with fewer than 50 employed individuals are excluded to avoid instability in the assessment caused by small sample sizes. Second, a stratified approach to missing data was applied: a threshold for the missing rate was first set at the occupational level to screen samples. If the missing rate for a key indicator in a given occupation exceeded 10%, it was deemed insufficient to support a reliable evaluation and was excluded. For occupations where the missing rate did not exceed the threshold, missing values were imputed using the indicator's mean within the same category and year. This approach minimizes sample size reduction while controlling noise, providing a consistent and reproducible data foundation for subsequent clustering analysis and comprehensive

Table 1: Overview of Data Sources and Their Roles in the Modeling Process

Data Source	Content Provided	Purpose in This Paper
BLS[6]	Occupational employment and wage statistics	Constructing labor demand and compensation return metrics
NCES[7]	Institutional attributes, enrollment, and outcome statistics	Characterizing institutional features and standardizing educational data connections
CSD[8]	Completion rates, debt, graduate earnings, etc.	Evaluating educational pathway returns and affordability
O*NET OnLine[9]	Occupational tasks, competencies, and technical skills	Occupational task-skill mapping and Gen-AI exposure proxy metrics

evaluation.

4 Notations

The model employs the following notations as shown in table 2.

Table 2: Notations

Symbols	Description	Unit
$\text{score}_i^{(t)}$	The composite score for occupation i in year t	—
$\bar{\mathbf{W}}$	The average dimension weight vector for the indicator system	—
K	The number of clusters	—
$\mathbf{R}_i$	Membership matrix for occupation or type i	—
$\mathbf{V}$	Influence level score vector	—
B_i	AI impact coefficient for occupation or type i	—
$G(t; k, t_0)$	Macro AI impact intensity function (Logistic)	—
k, t_0	Logistic parameters: acceleration rate and inflection year	year
$\mathbf{x}$	Institutional-level decision vector: course structure and enrollment adjustments	—
$f_1, \dots, f_6$	Six objective functions	—
λ_i	Objective weight coefficients	—
Z	Comprehensive objective function	—
l_i, h_i	Decision variable lower/upper bounds	—
α, p, β	Key parameters: coupling, nonlinear sensitivity, and discount factor	—
C_i	Unit cost coefficient	—
P_1, P_2	Constraint penalty terms	—

5 CLARITY: CLustering Assessment for foRecast optImiza-Tion sYstem

5.1 Sub-model I: Entropy-Weighted Clustering(EWC)

5.1.1 Indicator Selection

Multiple factors, including employment demand, compensation returns, skill structures, and levels of technology application, typically influence the development trajectories of different occupations in the labor market. To systematically characterize the temporal evolution of occupations and identify occupational types with similar development patterns, this paper constructs Submodel I, establishing a unified, comprehensive occupational evaluation system for multidimensional quantitative analysis of occupational development states. Here, we denote $i = 1, 2, \dots, m$ as the occupation index and $t \in \{t_1, t_2, \dots, t_n\}$ as the year index. For observed data of sub-occupation i in year t , this paper employs McKinsey Logic Tree Analysis. By sequentially decomposing occupational development factors, it constructs a quantitative representation system for occupational development. Specific dimensions and metric definitions are shown in Table 3.

Due to significant data gaps in 2025, we collected data from over 200 positions spanning 2015 to 2024 to assess industry development over the past decade and assigned category labels to each occupation. To standardize units, we transformed the negative indicators $LMD3_i^{(t)}$ and $CER3_i^{(t)}$, then normalized the data using the min-max method. Building on this, the normalized data served two core objectives: first, integrating multidimensional metrics into a single annual composite score to ensure comparability across occupations and time periods and second, selecting representative occupations from each category by combining their current development levels with clustering characteristics.

We employ a two-step Entropy Weighting Method(EWM): first, we calculate the score for each indicator within every dimension; then, we compute the weight for each dimension to obtain the overall score for the task. Assuming the $w^{(1)}$ denotes the weights of dimensions and the $w^{(2)}$ denotes the weights of indicators, the annual composite score for occupation i in year t can be defined as

$$\text{score}_i^{(t)} = \sum_{l=1}^4 w_l^{(1)} \sum_{j=1}^d w_j^{(2)} x_{ij}^{(t)}. \quad (1)$$

By connecting the composite scores for each year of a given profession in sequence, one can obtain the multi-year composite score trajectory vector:

$$\text{score}_i = [\text{score}_i^{(t_1)}, \text{score}_i^{(t_2)}, \dots, \text{score}_i^{(t_n)}]. \quad (2)$$

Table 3: Indicator System of Submodel I

Symbol	Indicator Definition	Unit
<i>Dim1: Labor Market Demand (LMD)</i>		
$LMD1_i^{(t)}$	Employment population size	–
$LMD2_i^{(t)}$	Employment growth rate	%
$LMD3_i^{(t)}$	Unemployment rate	%
<i>Dim2: Compensation and Economic Return (CER)</i>		
$CER1_i^{(t)}$	Median annual salary	USD/year
$CER2_i^{(t)}$	Salary growth rate	%
$CER3_i^{(t)}$	Income dispersion ratio	%
$CER4_i^{(t)}$	Education-return ratio	%
$CER5_i^{(t)}$	Middle-to-high income premium	%
<i>Dim3: Skill Structure (SKS)</i>		
$SKS1_i^{(t)}$	Share of high-skill positions	%
$SKS2_i^{(t)}$	Skill update frequency	1/year
<i>Dim4: Technology Dependence (TED)</i>		
$TED1_i^{(t)}$	Digital tool adoption rate	%
$TED2_i^{(t)}$	Automation potential index	–
$TED3_i^{(t)}$	Technology investment intensity	USD/year
$TED4_i^{(t)}$	Workflow standardization level	index

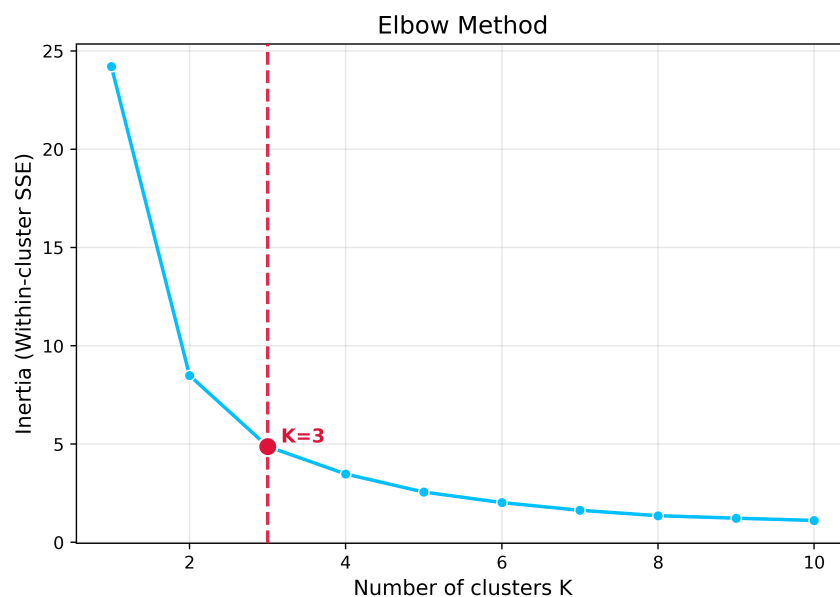


Figure 3: Elbow Method for Determining the Number of Clusters

To identify representative occupational types, we performed cluster analysis based on the similarity of recent occupational score trajectories. Specifically, we used $score_i$ as the feature vector for each occupation i and applied K-means clustering to the feature vector set $\{score_i\}$ for all occupations. This divided occupations into K groups, revealing distinct distributions for STEM, trade, and arts occupations. Finally, we selected the most representative occupations from each cluster by considering both the proportion of occupations within each cluster and the distance of each occupation from the cluster center. As shown in Figure 3, the elbow method determined the optimal number of clusters to be 3.



Figure 4: A two-dimensional t-SNE visualization of occupations colored by their occupational category labels (Trade, STEM, or Arts). The dashed circles in the figure represent clustering results.

5.1.2 Results: Type Explanation, Representative Occupations, and Trend Forecasts

To visually represent the clustering structure in the high-dimensional composite indicator space, this paper employs the t-SNE algorithm to map occupational samples onto a two-dimensional plane for visualization, as shown in Figure 4. The three distinct clusters of data points exhibit clear separation boundaries, indicating that within the feature space defined by composite indicators, different occupational types form relatively stable clustering structures rather than continuous distributions along a single dimension. Based on the clustering results, this paper selects the job within each cluster that is closest to the cluster center and belongs to an occupational category with a large proportion within the cluster group as the representative for each category. Ultimately, the

first cluster is represented by Railway Maintenance Worker (trade category), the second cluster by Management Analyst (STEM category), and the third cluster by Graphic Designer (arts category).

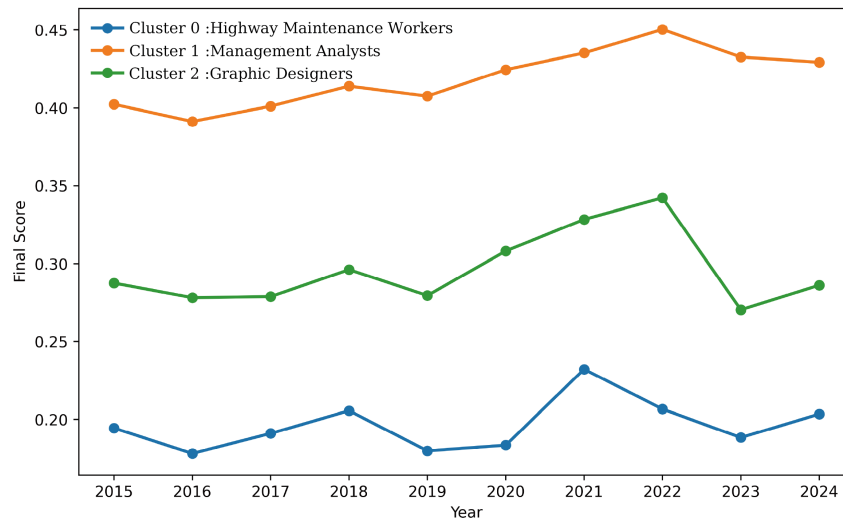


Figure 5: Annual Composite Score Trend Chart for Three Representative Occupations.

As shown in Figure 5, the composite scores for the three occupations exhibit stable and clearly tiered long-term trends, accompanied by normal annual fluctuations. Management analysts have maintained higher composite scores for many years, demonstrating stronger demand signals and superior educational returns. Railway maintenance workers show relatively low composite scores, with higher exposure to automation and slow skill updates. Graphic designers rank in the middle but face the most significant future impact from AI, with the greatest demand divergence between high-end and low-end roles.

In summary, Submodel I characterizes cross-occupation heterogeneity and selects three representative occupations—railway maintenance workers (Trade), management analysts (STEM), and graphic designers (Arts)—as anchors for subsequent analysis, upon which the follow-up models quantify the future impact of Gen-AI on career development and translate the projected shifts into actionable adjustments for higher-education training programs.

5.2 Sub-model II: Fuzzy Diffusion Timecasting (FDT)

5.2.1 Fuzzy Comprehensive Evaluation of Gen-AI Impact

The impact of Gen-AI varies significantly across different occupations. Some professions will experience rapid growth alongside AI development, while others will face relatively limited influence. Still others may face partial replacement by artificial intelligence. These shifts in career development patterns will directly influence talent cultivation programs within relevant disciplines at higher education institutions. To clarify future trends across occupations and provide a basis for higher education institutions to formulate talent development reform strategies for AI-impacted industries, this paper employs the Fuzzy Comprehensive Evaluation (FCE) method. Based on multidimensional foundational data from Submodel I, we conduct a quantitative assessment of

Table 4: Occupational Type: Explanatory Summary, and Representative Occupations.

Cluster	Interpretation	Code (Source: BLS[6])	Representative occupation
C0 (Trade)	Large workforce but slow growth; moderate educational returns; low-to-medium skill dependency with slow skill renewal. Routine, experience-driven jobs face higher substitution risks in generative AI contexts.	47–4051	Railway Maintenance Worker
C1 (STEM)	Rapid job growth, high compensation, strong educational returns, frequent digital tool usage, high technology investment intensity, and a high proportion of high-skill positions. Strong human capital and technology complementarity maintain a high composite score and a low risk of job displacement.	13–1111	Management Analyst
C2 (Arts)	Significant income volatility and pay disparities; divergent educational returns; uneven technology reliance with low automation risk; overall moderate score, but higher uncertainty and more pronounced tail risks.	27–1024	Graphic Designer

AI's impact across occupations along multiple dimensions. We derive the evaluation set from Submodel I.

$$U = \{\text{Dim1}, \text{Dim2}, \text{Dim3}, \text{Dim4}\}. \quad (3)$$

Similarly, we adopt the annual weights from Submodel I to obtain the historical weight set.

$$W^{(t)} = [w_1^{(t)}, w_2^{(t)}, w_3^{(t)}, w_4^{(t)}]. \quad (4)$$

We have established five possible evaluation outcomes.

$$V = \{\text{Strong Positive Impact}, \text{Moderate Positive Impact}, \text{Neutral Impact}, \text{Moderate Negative Impact}, \text{Strong Negative Impact}\}. \quad (5)$$

Assign them scores from the set $\{0.2, 0.1, 0, -0.1, -0.2\}$, enabling us to quantify the overall impact. Simultaneously, define the midpoints of each grade as $\{0.08, 0.04, 0, -0.04, -0.08\}$.

To reduce the subjective nature of the membership matrix R , which relies entirely on manual settings, we utilize research [10, 11, 12, 13, 14] on future work changes Using Natural Language Processing (NLP) technology to extract keywords from literature, and comparing them against the keywords we set for each dimension, we measure the impact on dimension j of occupation i

$$y_{ij} = \frac{\text{positive_word} - \text{negative_word}}{\text{total_word}}. \quad (6)$$

The numerator represents the difference between positive and negative keywords, measuring whether Gen-AI exerts a positive or negative influence. The denominator accounts for variations in corpus length. Meanwhile, we employ the y_{ij} mapping to reflect the membership matrix of AI's impact across various dimensions. Ultimately, by analyzing the dominant keywords across each dimension, four core driving factors are distilled to explain the mechanisms behind the projected changes: *AIGC Productivity Expansion*, *Value Redistribution Effect*, *Skill Reconstruction & Human-AI Collaboration*, and *Task Automability*; these drivers provide an interpretable bridge between the qualitative evidence base and the quantitative impact coefficient used in Submodel II.

Based on existing research and evaluation results, we use the triangular membership function

$$r_{jd}^{(i)} = \max\left(1 - \frac{|y_{ij} - c_d|}{d_0}, 0\right) \quad (7)$$

to map text data onto the membership matrix $\mathbf{R}_i \in [0, 1]^{4 \times 5}$. Here, $r_{jd}^{(i)}$ represents the probability that the AI influence on dimension j of occupation i falls within level d . c_d denotes the level center, and d_0 is the half-width, set to 0.4.

Based on the NLP results, membership functions, and manually fine-tuned outputs, we visualize the membership patterns for the Trade, STEM, and Arts occupational categories using heatmaps (Figure 6). The color intensity directly reflects the membership probability for each dimension-level pair, providing an interpretable comparison across categories.

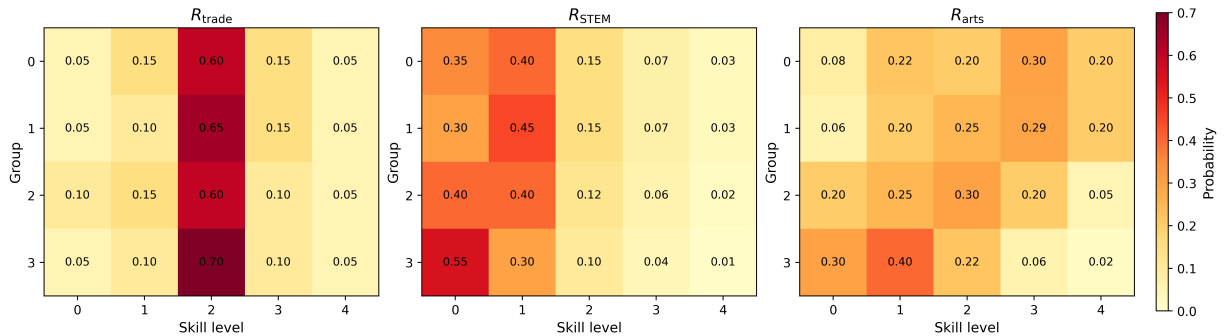


Figure 6: Membership heatmaps for Trade, STEM, and Arts categories. Color indicates membership probability for each dimension (row) and impact level (column).

Our analysis of the membership matrix $\mathbf{R}$ reveals that driving factors exert varying effects on the same dimension across different occupations. For Labor Market Demand, the driving logic lies in enhancing the production efficiency of content and services, indirectly leading to changes in industry demand scale and structure. In trade occupations, AI has a limited impact on core tasks such as railway maintenance and line upgrades. Consequently, 60% of the probabilities in the membership matrix cluster around neutral influence, indicating a gradual driving effect. In STEM occupations, AI significantly boosts efficiency in core tasks such as data analysis and report generation, resulting in a 75% positive rating in the attribution matrix. For arts occupations, AI efficiently produces basic posters and template-based designs, squeezing the market and causing 50% of indicators in the attribution matrix to be negative.

In the Compensation and Economic Return dimension, Gen-AI reshapes the value density of occupational tasks, adjusting market pricing logic for different skills. For Trade occupations, AI has not significantly altered the value structure of core tasks, leaving wage growth lacking strong momentum. Consequently, 65% of outcomes fall within the neutral impact category, with modest return-driven effects. However, within STEM professions, the deep integration of AI with business analytics substantially enhances decision-making efficiency and commercial value. Practitioners proficient in AI tools can generate higher marginal benefits, resulting in a positive evaluation rate of up to 75%. In Arts professions, AI replaces a significant volume of foundational design tasks, diluting the labor value of low-skilled practitioners and causing wage levels to decline, with negative impacts approaching 50%.

Within the Skill Structure, Gen-AI automates foundational tasks, compelling practitioners to transition toward advanced skills that AI cannot replace. In Trade occupations, core competencies remain centered on complex troubleshooting an area AI cannot replace resulting in a 60% probability of neutral impact. Within STEM professions, AI automates low-value tasks like data cleaning and basic modeling, compelling practitioners to shift toward advanced skills such as model optimization, business interpretation, and AI tool calibration. These new competencies exhibit intense scarcity and high adaptability, contributing to an 80% positive impact in the matrix. In the Arts professions, AI replaces foundational skills such as basic drawing and color matching. Practitioners must transition toward original creative planning and the optimization of AI-generated content. However, some entrepreneurs struggle to adapt due to creative constraints, leading 55% of respondents to report neutral or negative assessments.

Within the Technology Dependence dimension, Gen-AI-driven occupations enhance automation potential and reliance on digital tools. Trade occupations, characterized by a strong hands-on nature and high uncertainty, remain concentrated in the neutral impact category with a 70% probability. STEM occupations, involving highly standardized, data-driven tasks such as data analysis and predictive modeling, exhibit strong AI automation feasibility, resulting in an 85% positive impact share. In the Arts professions, AI painting tools and creative generation platforms significantly boost productivity. Basic creative tasks are highly automatable, driving increased reliance on AI tools among practitioners. Positive impacts account for 70%, with technology dependence demonstrating a pronounced driving effect.

5.2.2 AI Impact Function Fitting and Parameter Calibration

The impact of AI on occupations is linked to the pace of AI technological advancement and adoption. Considering that the diffusion of generative AI typically follows a phased pattern, we employ a Logistic diffusion function to characterize the temporal evolution of macro-level AI impact intensity:

$$G(t; k, t_0) = \frac{1}{1 + \exp[-k(t - t_0)]}. \quad (8)$$

Here, t denotes the calendar year, k represents the diffusion acceleration rate, and t_0 is the inflection point year. Based on relevant research[15], we fitted the parameters to $k = 0.8$ and $t_0 = 2027$.

5.2.3 Time Series Models Predict Future Career Development

In Submodel I, different occupations exhibit distinct but relatively stable trajectories. Here, we combine baseline trend estimation with AI influence to forecast future trajectories. Submodel II employs smoothing time-series analysis to estimate the baseline trend of $score_i$ over the last 10 years. The time series forecasting steps are as follows:

(1) Single Exponential Smoothing. Based on Submodel I, let $S_i^{(t)}$ denote the composite score for occupation i in year t . The recursive form is

$$\hat{S}_i^{(t)} = \alpha S_i^{(t-1)} + (1 - \alpha) \hat{S}_i^{(t-1)}, \quad 0 < \alpha < 1. \quad (9)$$

Where $\hat{S}_i^{(t)}$ represents the single exponential smoothing estimate for occupation i in period t , $S_i^{(t)}$ denotes the actual observed value for occupation i in period t , and $\hat{S}_i^{(t-1)}$ is the single exponential smoothing estimate for occupation i in period $t-1$; α is the exponential smoothing coefficient, with values ranging from $0 < \alpha < 1$. It controls the weight allocation between the current actual observation and the previous smoothed estimate in the current smoothing result. The magnitude of the smoothing coefficient α directly determines the weight allocation: a larger α increases the weight of the current actual observation, making the smoothed result more responsive to recent data changes; a smaller α increases the weight of the previous smoothed estimate, enhancing the overall stability of the smoothed result. To avoid subjective setting of the smoothing coefficient, this paper defines a candidate set $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ and selects values based on back-testing error: using $t \in [2015, 2019]$ as the training segment to obtain $\hat{S}_i^{(2019)}$, which serves as the baseline anchor for the validation segment

$$\text{baseline}_i = \hat{S}_i^{(2019)}, \quad (10)$$

and compute the mean absolute error over $t \in [2020, 2024]$:

$$\text{MAE}(\alpha) = \frac{1}{5} \sum_{t=2020}^{2024} \left| S_i^{(t)} - \text{baseline}_i \right|, \quad (11)$$

Select $\alpha = 0.5$ that minimizes $\text{MAE}(\alpha)$ as the optimal smoothing coefficient. After obtaining the optimal α , this paper re-smooths the entire historical interval $t \in [2015, 2024]$ to form a forecast baseline, and uses

$$\text{baseline}_i = \hat{S}_i^{(2024)}$$

as the non-AI-impacted benchmark for future forecasts.

(2) Weight stabilization. To mitigate interannual fluctuations, we employ the average weights:

$$\bar{\mathbf{W}} = \frac{1}{10} \sum_{t=2015}^{2024} \mathbf{W}^{(t)}. \quad (12)$$

Accordingly, the AI impact magnitude for occupation i is defined as

$$B_i = \bar{\mathbf{W}} \mathbf{R}_i \mathbf{V}. \quad (13)$$

(3) AI-adjusted forecast. Combining the macro-shock intensity $G(t; k, t_0)$, the AI-adjusted forecast is

$$S_{i,\text{future}}^{(t)} = \underbrace{\text{baseline}_i}_{\text{No AI baseline}} + \underbrace{B_i G(t; k, t_0)}_{\text{AI adjustment}}. \quad (14)$$

When AI is in its early diffusion phase, $G(\tau; k, t_0)$ is small, and the predicted values approach the baseline without AI. As AI approaches the mature diffusion phase, $G(\tau; k, t_0) \rightarrow 1$, and the prediction offset is primarily determined by the sign and magnitude of B_{ij} . Consequently, the model can characterize both net enhancement and net substitution pathways for different occupations within the same framework, producing standardized results suitable for scenario analysis and sensitivity analysis.

(4) Initialization settings. To ensure computational feasibility of the recursion, the smoothing sequence is initialized with $S_j^{(2015)} = S_j^{(2015)}$. This initialization remains robust under limited annual samples and avoids introducing additional hard-to-interpret free parameters.

5.2.4 Results: AI-adjusted Trajectories for Trade, STEM, and Arts

Figure 7 compares the observed composite scores (2015–2024) with the AI-adjusted forecasts produced by (14). Overall, the predicted curves remain within the historical fluctuation band and preserve the relative ranking across the three categories, indicating that the baseline smoothing and the AI adjustment term jointly generate stable and interpretable trajectories.

For **Trade**, the observed series shows moderate year-to-year variability, including a pronounced transient spike around 2021. The forecast smooths this outlier and projects a mild upward drift through the late 2020s, followed by a stabilization near the historical mean. This pattern suggests that Gen-AI acts primarily as a *limited modifier* for trade occupations, with the net effect dominated by baseline labor-market fundamentals rather than a strong AI-driven trend.

For **STEM**, both the historical observations and the forecast exhibit a clear upward trajectory. After 2024, the predicted path accelerates and reaches a higher plateau in the early 2030s, consistent with a positive impact coefficient and a diffusion process approaching maturity. In practical terms, STEM occupations are more likely to benefit from Gen-AI through productivity gains and task complementarity, yielding a sustained increase in the composite score.

For **Arts**, the observed series contains a sharp rise before 2022 and a notable drop thereafter, reflecting higher volatility and structural uncertainty. The forecast does not extrapolate the pre-2022 surge; instead, it converges to a relatively flat path with small oscillations, implying that the net AI effect is mixed: high-end creative roles may be complemented, while routine content production is more exposed to substitution. As a result, the long-run composite score for arts-related occupations is projected to remain stable but with elevated downside risk relative to the other categories.

5.3 Sub-model III: PSO Multiobjective Allocation (PMA)

At the institutional level, this paper selects one representative program from each of three educational pathways: Railway Engineering Technology at Vocational College A, Business Analytics at Comprehensive University B, and Visual Communication Design at Art College C.

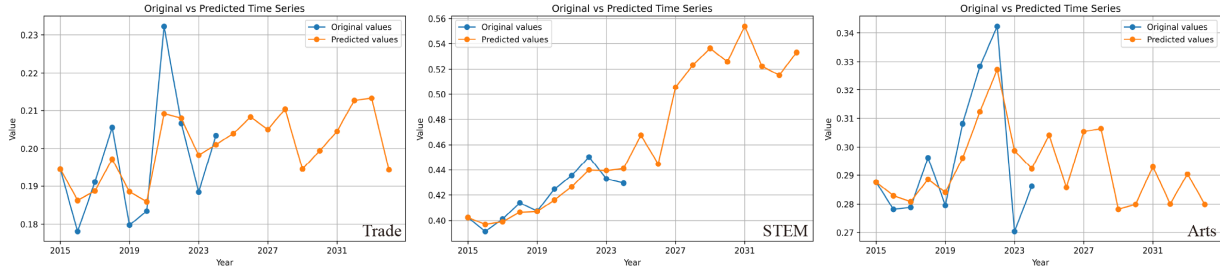


Figure 7: Original (2015–2024) versus AI-adjusted predicted composite scores for the representative Trade, STEM, and Arts occupations.

5.3.1 Multi-objective optimization

Objective 1: Maximize graduate employability.

To measure the overall alignment between the program and the labor market, we constructed an employability indicator based on quantifiable training elements selected from four dimensions of Submodel I: labor market demand, economic returns, skill structure, and technology dependency. These indicators are: proportion of industry-academia collaboration courses x_1 , proportion of core courses x_2 , proportion of skill certification training courses x_3 , and proportion of practical courses x_4 , weighted by Dim1 to Dim4. The expression for Objective 1 is as follows:

$$f_1 = \sum_{i=1}^4 w_i x_i. \quad (15)$$

Objective 2: Optimize program enrollment scale.

To align enrollment scale with labor demand, this paper employs the employment growth rate indicator LMD2 from Submodel I as the market demand signal. Simultaneously, indicator x_5 is defined as the rate of change in enrollment scale for the major. Intuitively, the closer x_5 is to LMD2, the lower the risk of supply-demand mismatch. Therefore, we establish the expression for Objective 2:

$$f_2 = -|x_5 - \text{LMD2}|. \quad (16)$$

Objective 3: Balancing learning efficiency and outcomes.

To simulate the double-edged effect of digital tools on the learning process, this paper adopts the digital tool usage rate indicator TED1 from Model I as a reference for reasonable tool dependency levels. Let x_6 denote the project's Gen-AI usage intensity, which can be measured by AIGC intervention rate or AI usage frequency in assignments. Excessively high x_6 may lead to superficial learning and inflated skill levels, while excessively low values may reduce learning efficiency and tool literacy. Therefore, this paper characterizes the principle of moderate use through a deviation penalty mechanism.

$$f_3 = -(x_6 - \text{TED1})^2. \quad (17)$$

Objective 4: Enhance knowledge accessibility and resource utilization.

For programs forced to reduce enrollment due to Gen-AI disruption, compensating for resource shortfalls can be achieved by expanding public general education courses. These courses, accessible to the broader public, both disseminate knowledge and generate revenue through moderate fees. This paper uses the proportion of public general education courses x_7 as an indicator, characterizing this objective through a positive utility mechanism:

$$f_4 = x_7 [\max(-x_5, 0)]^p. \quad (18)$$

Objective 5: Control ethical risks and ensure originality.

Intuitively, increased AI usage intensity heightens risks of plagiarism, unclear authorship, and ethical violations. We define x_8 as the proportion of AI ethics and original content training courses, aiming to mitigate these risks through standardized training while enhancing originality awareness. To simultaneously reflect the positive effects of training investment and the negative effects of high-intensity AI usage, this paper defines Objective Five as

$$f_5 = x_8(1 - x_6). \quad (19)$$

Objective 6: Reduce waste of social resources.

AI usage incurs additional energy and water consumption. We characterize resource conservation effects through the linear term of x_6 , introducing a discount factor β to represent its relative weight. Objective 6 is expressed as: While we encourage students to consider environmental factors when using AI, this should not become the primary constraint on AI adoption. Therefore, β is set to 0.3. This modest value reflects AI's efficiency gains while avoiding excessive suppression of educational innovation.

To uniformly balance the six objectives, the model adopts the weighted-sum method to construct an overall utility function:

$$\max Z = \sum_{i=1}^6 \lambda_i f_i. \quad (20)$$

The weight of the primary objective f_1 is set to $\lambda_1 = 0.5$, while the remaining weights are set to 0.1.

The constraints are defined as

$$\text{s.t.} \quad \begin{cases} l_i \leq x_i \leq h_i, & i = 1, 2, \dots, 8, \\ x_1 + x_2 + x_3 + x_4 + x_7 + x_8 \leq 1, \\ C_1x_1 + C_2x_2 + C_3x_3 + C_4x_4 + C_7x_7 + C_8x_8 \leq C_0. \end{cases} \quad (21)$$

Here, the first constraint enforces the lower and upper bounds of each decision variable; the second constraint ensures the feasibility of the curriculum structure through a proportion constraint; and the third constraint defines the budget feasibility region by restricting the total cost of all course types not to exceed the overall budget. Regarding the bound constraints, for variables $x_1 \sim x_6$, constant lower and upper limits are specified according to common practice. Since program expansion crowds out resources allocated to general education courses, the upper bound of x_7 is

designed to depend on the scale of enrollment expansion. For x_8 , because AI risk is associated with the intensity of AI usage, the institution should guarantee a minimum investment in AI ethics courses; thus, its lower bound is linked to AI usage intensity. Based on the above considerations, the variable bounds for three programs at three institutions are specified as follows:

(1) Trade school A Railroad Engineering Technology:

$$\mathbf{l}^{(A)} = (0.05, 0.2, 0.05, 0.05, -0.25, 0, 0, \alpha x_6) \quad (22)$$

$$\mathbf{h}^{(A)} = (0.5, 1, 0.5, 0.5, 0.25, 1, 0.5 - x_7[\max(0, x_5)]^p, 0.3) \quad (23)$$

(2) University B Business Analytics:

$$\mathbf{l}^{(B)} = (0, 0.5, 0, 0, -0.25, 0, 0, \alpha x_6) \quad (24)$$

$$\mathbf{h}^{(B)} = (0.3, 1, 0.3, 0.3, 0.25, 1, 0.5 - x_7[\max(0, x_5)]^p, 0.15) \quad (25)$$

(3) Arts school C Graphic Design:

$$\mathbf{l}^{(C)} = (0.05, 0.3, 0.05, 0.05, -0.25, 0, 0, \alpha x_6) \quad (26)$$

$$\mathbf{h}^{(C)} = (0.3, 1, 0.5, 0.5, 0.25, 1, 1 - x_7[\max(0, x_5)]^p, 0.3) \quad (27)$$

where α is the coefficient controlling the impact of AI intensity on AI ethics courses and is set to 0.3, and p measures the effect of enrollment expansion on general education courses and is set to 1.5.

For the third constraint, C_i denotes the unit cost corresponding to course category x_i , and C_0 is the total annual budget of the program. The total program budget depends on enrollment scale and the current status of the occupation. The cost vector is specified as

$$\mathbf{C} = (15, 5, 10, 10, -2, 1), \quad (28)$$

and after normalizing the baseline in Model II, the adjusted total budget regulated by enrollment scale and industry conditions is computed as

$$C_0 = 20(1 + x_5) \text{ baseline}. \quad (29)$$

5.3.2 PSO for Solving the Curriculum Allocation Problem

To solve the aforementioned optimization model in continuous space, this paper employs a particle swarm optimization algorithm to search for the optimal course allocation ratio. Each particle represents a feasible solution. $\mathbf{X} = (x_1, \dots, x_8)$, and use the composite objective value Z as the fitness function:

$$\text{fitness}(\mathbf{X}) = Z \quad (30)$$

Where $\eta = 10$. For the budget constraint, define the penalty term

$$P_2 = \max(C_1x_1 + C_2x_2 + C_3x_3 + C_4x_4 + C_7x_7 + C_8x_8 - C_0, 0) \quad (31)$$

Thus, the objective function actually optimized by PSO is

$$\max \text{fitness}(\mathbf{X}) - P_1 - P_2. \quad (32)$$

By examining the characteristics of the three selected educational pathways, we initialize PSO with three program-level baseline solutions. The three programs correspond to:

- (i) a Business Analytics program at an East-coast top-tier engineering-focused institution (STEM)
- (ii) a Railway Engineering Technology program at a Midwest applied-technology college (Trade)
- (iii) a Graphic Design program at a well-known New England art-focused institution (Arts)

The institution names are intentionally anonymized, and all qualitative program attributes used for setting initialization are derived from publicly available program descriptions and curriculum information.

Accordingly, the initial solutions are set as

$$\mathbf{X}^{(A)} = (0.16, 0.32, 0.14, 0.17, 0.06, 0.30, 0.10, 0.11),$$

$$\mathbf{X}^{(B)} = (0.10, 0.55, 0.05, 0.05, 0.13, 0.70, 0.15, 0.10),$$

$$\mathbf{X}^{(C)} = (0.10, 0.35, 0.08, 0.22, -0.08, 0.50, 0.15, 0.10).$$

We set the swarm size to $N = 100$, the maximum number of iterations to $K = 200$, the individual learning factor to $c_1 = 2$, and the social learning factor to $c_2 = 2$. The inertia weight at step k is $\omega^{(k)} = \omega_{\max} - \frac{\omega_{\max} - \omega_{\min}}{K} k$. Here $\omega_{\max} = 0.9$ and $\omega_{\min} = 0.4$, indicating that the inertia weight decays linearly as the iteration index increases, so the swarm transitions from an exploration mode in the early stage to a more conservative mode in the later stage.

The algorithm stops when the maximum number of iterations K is reached and outputs the optimal solution $\mathbf{X}^*$. The updated utility value $\mathbf{Z}^*$ is then obtained from the optimal solution. The improvement of the proposed scheme is quantified by the relative gain:

$$\text{Improvement} = \frac{Z^* - Z}{Z}. \quad (33)$$

To further interpret the multi-objective trade-offs behind the aggregate gain, Figure 8 visualizes the normalized objective achievements before optimization and after optimization for the Trade, STEM, and Arts programs. Each radar axis corresponds to one objective f_1 – f_6 . The expansion of the blue polygon relative to the red polygon indicates that PSO improves the overall objective balance rather than optimizing a single metric in isolation.

Across all three programs, the optimized solutions consistently strengthen the core performance axes f_2 and f_3 , suggesting that the recommended curriculum allocation prioritizes alignment with demand and learning effectiveness under Gen-AI-assisted training. In addition, the Trade and Arts programs show a substantial improvement on the environmental/resource-related objective f_6 , while maintaining competitive performance on employment-related objectives. For

the STEM program, the optimization yields a particularly pronounced gain on f_1 and f_2 , consistent with a strategy that scales market-facing and analytics-intensive components while keeping ethical and resource objectives within feasible bounds. Overall, the radar comparison confirms that the optimized curriculum packages are not only higher in total utility Z^* but also more structurally balanced across the six objectives, supporting actionable recommendations for institutional adjustment.

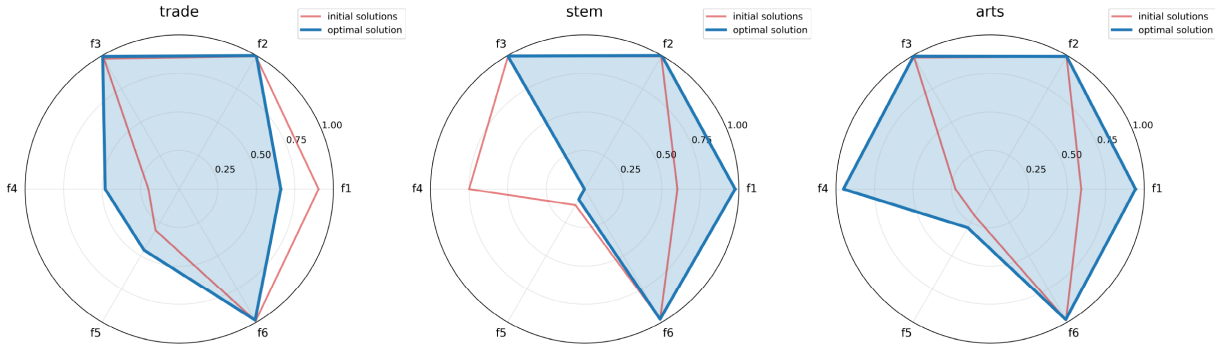


Figure 8: Radar comparison of normalized objective achievements (f_1 – f_6) between initial solutions and PSO-optimal solutions for the Trade, STEM, and Arts programs.

5.3.3 Policy Recommendations

Table 5: University mission profiles and objective weight vectors.

Profile	Weights λ	Interpretation
(A)	(0.25, 0.10, 0.30, 0.05, 0.15, 0.15)	Research-oriented universities focus on cutting-edge innovation and resource conservation.
(B)	(0.15, 0.10, 0.10, 0.15, 0.25, 0.25)	Ethics- and eco-conscious universities emphasize restricting AI usage, strengthening governance, and promoting sustainable development.
(C)	(0.15, 0.20, 0.05, 0.30, 0.15, 0.15)	Commercialized universities prioritize market sensitivity and revenue generation while meeting compliance requirements.

Based on the above table 5 and the radar-chart results, the following recommendations are proposed:

Trade School A – Railway Maintenance Technology

Increase general education courses to 20%

Expanding general education strengthens interdisciplinary knowledge and improves the accessibility and public reach of technical training.

Increase AI ethics training to about 20%

Enhancing AI ethics education reduces operational risks and prepares students to responsibly use AI-assisted maintenance technologies.

Reduce routine technical and practice courses to about 10%

Since this occupation is less vulnerable to AI substitution, reducing repetitive technical training allows resources to shift toward broader competency development.

Expected Outcome:

Training quality remains stable while improving professional outreach and AI risk management.

University B – Business Analytics Program

Increase industry–university collaborative courses to 25%

Stronger industry partnerships improve practical skills and better align training with labor market needs.

Control enrollment expansion below 10%

Slowing expansion reduces talent oversupply and keeps program growth consistent with AI-driven labor demand.

Temporarily suspend general education courses

Redirecting teaching resources toward professional courses helps maintain training quality during rapid program expansion.

Expected Outcome:

Overall program performance is projected to increase significantly, with substantial improvements in graduate employability.

Arts School C – Graphic Design Program

Reduce traditional practice courses to about 5%

As AI replaces routine design tasks, the curriculum should focus more on advanced creative and conceptual skills.

Increase AI tool training intensity to about 60%

AI-assisted design skills are essential for future industry competitiveness.

Maintain general education courses at about 10%

This balance supports educational outreach while maintaining core training quality.

Expected Outcome:

Graduate employability and AI competency improve notably, while expanding access to art education. By analyzing relevant data from other institutions, our model can be further extended to more universities and programs within the same occupational category. This will help them develop more rational curriculum designs and enrollment plans to address the challenges and opportunities presented by generative AI.

6 Sensitivity Analysis

This section examines the robustness of Model III to variations in the university preference weights λ . The baseline setting prioritizes employment as the core objective. Compared to this baseline, the missions and constraints of different types of universities exhibit systematic differences, potentially making the optimal policy solution $\mathbf{x}^*$ sensitive to λ . To transform this issue from a scenario description into a repeatable sensitivity experiment, this paper employs a two-stage design combining discrete profiling with local perturbations. First, three mission profiles are constructed as representative weighting points. Then, the weights near each profile are perturbed by $\pm 10\%$.

6.1 Objectives

Let the normalization target vector be

$$\mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), f_3(\mathbf{x}), f_4(\mathbf{x}), f_5(\mathbf{x}), f_6(\mathbf{x}))^\top. \quad (34)$$

Comprehensive utility is

$$Z(\mathbf{x}; \lambda) = \sum_{k=1}^6 \lambda_k f_k(\mathbf{x}), \quad \lambda_k \geq 0, \quad \sum_{k=1}^6 \lambda_k = 1, \quad (35)$$

Here, $\mathbf{x}$ represents policy variables, while $f_k(\cdot)$ corresponds to the six objectives in Q3. This paper constructs three types of university profiles and assigns corresponding weights as follows:

By separately solving for the optimal policy solutions $\mathbf{x}^{*(A)}$, $\mathbf{x}^{*(B)}$, and $\mathbf{x}^{*(C)}$, we obtain the corresponding objective achievement vectors $\mathbf{F}(\mathbf{x}^{*(\cdot)})$. These vectors are used to compare how mission differences reshape the internal policy structure at the macro level. As shown in Figure 9, the three preference profiles generate distinct trade-off patterns across the six objectives. In particular, the research-oriented profile allocates more emphasis to performance- and innovation-related goals, while compressing objectives associated with governance or externalities. The ethics- and eco-conscious profile shifts the solution toward risk control and sustainability, yielding a more conservative structure with improved achievements on the compliance- and resource-related axes but weaker performance on market-expansion and revenue-driven objectives. The commercialized profile exhibits the most pronounced reallocation toward market sensitivity and scalability, increasing achievements on growth-aligned objectives while accepting larger sacrifices on ethical-risk mitigation. Overall, the radar comparison provides a transparent, category-consistent diagnostic that preference heterogeneity is a primary driver of the optimal curriculum and enrollment policy mix, thereby motivating the subsequent local perturbation analysis.

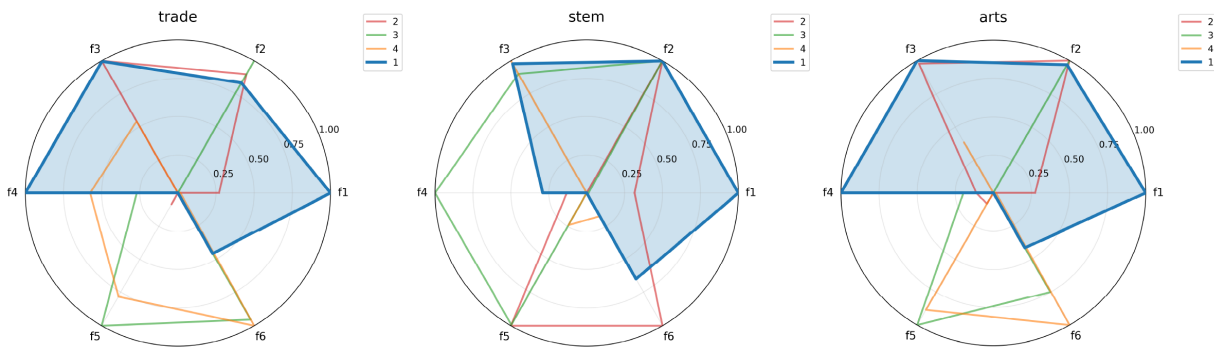


Figure 9: Radar comparison of normalized objective achievements under different preference profiles for Trade, STEM, and Arts programs.

7 Strengths and Weaknesses

7.1 Strengths

1. Traceability and Enforceability

The model establishes a process encompassing occupational identification, impact quantification, and solution optimization. It first screens representative occupations from the data, then clarifies the direction and intensity of generative AI's impact on each occupation. Finally, within budgetary and structural constraints, it outputs actionable plans for curriculum design and enrollment arrangements, ensuring verifiable conclusions and implementable recommendations.

2. Interpretability and Scalability

The model integrates established methodologies, including composite scoring, clustering typology, fuzzy evaluation, and multi-objective optimization, featuring intuitive parameter settings with clear definitions. Its outputs correspond to multiple practical dimensions: labor demand, return on investment, skill structure, and technology dependency, demonstrating strong interpretability and comparability. For countries at varying stages of AI development, the same modeling process can be reapplied to other occupations or educational programs by simply adjusting the data inputs and relevant parameters.

3. Robustness and Adaptability

By conducting sensitivity experiments that combine institutional mission profiling, the stability of the preference-setting approach is validated. Concurrently, a constraint penalty mechanism ensures solutions remain feasible within realistic boundaries. This enables tiered policy packages tailored to different institutional types, enhancing the model's transferability and implementation effectiveness.

7.2 Weaknesses

1. Aggregation bias and insufficient heterogeneity

Applying a single impact coefficient to each representative occupation may overlook differ-

ences in job specialization, variations in technological adoption within organizations, and disparities in task reorganization within the same occupation. Therefore, the model is better suited for relative comparisons and trend analysis across different occupational types, but struggles to provide precise predictions for specific job positions.

2. Parameter Dependency and Context Sensitivity

Key parameters of the diffusion function, such as inflection points and acceleration rates, influence the timeframe within which forecasted changes manifest. Membership mappings also depend on the corpus and rule configurations, so results should be presented alongside baseline scenarios and sensitivity analyses to avoid interpreting any single development path as absolute.

8 Conclusion

This paper proposes a closed-loop framework for institutional decision-making that encompasses career trend identification, quantification of Gen-AI impact, and curriculum optimization. Using one representative occupation from each of the three major categories STEM, Trade, and Arts as the primary analytical thread, the framework implements three sub-models based on four major public databases (BLS, NCES, College Scorecard, O*NET). Sub-model I constructs a multidimensional indicator system, calculates annual composite scores for occupations using entropy weighting, and performs K-means clustering based on score trajectories to identify stable occupational types and determine representative occupations. Sub-model II employs fuzzy comprehensive evaluation to generate membership matrices and standardized AI impact coefficients. Combined with a logistic diffusion function and an exponential smoothing baseline, it enables interpretable predictions of future occupational development. Submodel III constructs a multi-objective optimization model for institutions. Using curriculum and enrollment adjustments as decision variables, it solves for optimal training plans under budget and feasibility constraints via particle swarm optimization. The integrated process translates career shifts into actionable policy packages for institutions. It tests the robustness and transferability of optimization recommendations across different institutional missions by perturbing preference weights.

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Report on Use of AI

This section provides transparent disclosure regarding the use of generative AI tools during the writing and figure creation processes of this paper. It also outlines manual verification and control measures to ensure that model conclusions and data results are traceable, reproducible, and auditable.

1. Chart Visual Element Generation.

Tools and Versions: OpenAI ChatGPT (GPT-5.2 Thinking).

Usage Location: Figure 1 (Gen-AI Impact Evidence).

Purpose: Used solely for stitching together diagrams of different AI models and products to form a circular pattern.

2. Translate the English question stem and literature into Chinese.

Tools and Versions: OpenAI ChatGPT (GPT-5.2 Thinking).

Purpose: Translate the English question stem and key paragraphs from English references into Chinese to help the team quickly align on the question's intent during the early modeling phase, clarify task decomposition, and organize the core conclusions, assumptions, and scope of application from the literature. This facilitates internal discussions and the selection of solutions.

Boundary: AI is solely used for language conversion and key point summarization. It shall not add new requirements, introduce information absent from the original text, or replace final verification of the English source material. All task descriptions and literature citations included in the paper are based on the English original text, with the team cross-checking each item to prevent misinterpretation of questions or misattribution of references due to translation discrepancies.

3. Suggestions for debugging code.

Tools and Versions: OpenAI ChatGPT (GPT-5.2 Thinking).

Purpose: Troubleshooting steps for local Python environment errors, along with suggestions for reusable plot template code.

Boundary: All numerical results, charts, and experimental outputs in this document were obtained by the team running experiments in a local environment. These were validated through repeated runs and parameter verification. AI did not directly generate any numerical results used for conclusions.

Manual Verification and Compliance Statement:

- (1) All core conclusions are derived solely from model calculations and data processing results;
- (2) Numerical values, labels, and meanings appearing in charts and graphs are manually verified item by item;
- (3) Any AI-generated text undergoes consistency checks against symbol tables, formulas, and parameter settings prior to integration;
- (4) AI is not used to fabricate data, forge citations, or substitute model validation procedures.